

BASAVARAJESHWARI GROUP OF INSTITUTIONS
BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT
Autonomous Institute under VTU, Belagavi
NAAC Accredited Institution*



(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated
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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

A Project Report on

“Shopping cart with auto billing system”

A dissertation submitted in partial fulfillment of the requirements for the
award of degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi, Karnataka

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CERTIFICATE

Certified that the project work entitled **“Shopping cart with auto billing system”**, is a bonafide work carried out by **Aishwarya P, B Neha Yadav, Divya M and Divyashree Mullur** bearing USN **3BR21EC008, 3BR21EC022, 3BR21EC041 and 3BR21EC042** the bonafide students of Ballari Institute of Technology and Management in partial fulfilment for the award of degree in **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visveswaraya Technological University, Belagavi** during the academic year 2024-2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

Signature of Guide

(Dr. William Thomas H M)

Signature of HOD

External Viva

Name of the Examiner

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Signature with Date

Chapter 1

INTRODUCTION

Presently, the shopping system used in the shopping malls is the Barcode System. This system has replaced the previous manual system but has limitations. To begin with, barcode system requires the barcode on the products to be in the line of site of the barcode scanner. Its scanning range is just from a few inches to a few feet. A barcode scanner can read products only one at a time. Barcodes define the type of every product but can't do it uniquely. Barcodes are read only type and can't be overwritten. The barcode system runs on optical (laser) technology. Barcodes also require a considerable amount of man power and human effort. Barcodes can get damaged easily. Not only this, the current Barcode system requires the customer to stand in long queues in order to get their products scanned and their bills generated. This process can prove to be tiresome and it also consumes a lot of time of the customers, thereby adding to their frustration. With so many disadvantages to it, Barcode system is still in use. It is obvious that there is a need to bring on a smarter and a more efficient system Smart cart using Arduino and RFID is a new advancement in the field of Supply Chain Optimization. This system shall not only eradicate the long queues in supermarkets and malls but also save a lot of time for the customers. The system also helps the customer in money management. The system uses RFID tags in the place of Barcode tags which are much more efficient and powerful when it comes to scanning of products. The device developed using Arduino and RFID shall be installed on the shopping cart or shopping basket and the customer shall scan their products themselves and the total generation shall happen on the cart itself. This shall also give an idea to the customers on how much their particular shopping session shall cost them. Hence, time management and money management, both shall be taken care of. In essence, this project is an embedded system. Therefore, this technology is used in supermarkets because it will reduce the need for staff while improving the shopping experience for customers. This shopping trolley assists the customer in cutting down on their shopping time and avoiding the lengthy line at the billing part. Numerous technologies are utilized in the Shopping Trolley. This Smart shopping cart mainly focuses on the user interaction with the cart during shopping. In the market there are no existing solutions like the shopping cart which displays purchased items in real-time.

PROBLEM STATEMENT:

When we visit a supermarket for shopping, we spend enough time in selecting the right product, after that we spend even more time in standing in queue for billing at the checkout counters. In the case of weekends or during festival seasons when everyone come to shop, we have to wait even more due to the heavy crowd. To eliminate waiting time and have greater interaction during shopping we need a smart shopping cart to eliminate all these problems.

OBJECTIVE OF PROJECT:

The main objective of this project is to design a “**Smart Shopping Cart**” various problems.

1. Update web-app when item is inserted or removed from the cart.
2. Display cart items in real time.
3. Automatic billing when button is pressed.

Chapter 2

LITERATURE SURVEY

- 1) Kumar, A. Gupta, S. Balamurugan, S. Balaji and R. Marimuthu, "Smart Shopping Trolley," 2017 International conference on Microelectronic Devices, Circuits and Systems (ICMDCS), 2017, pp. 1-4. This paper proposes a system that reduces queues at billing counters in shopping complexes by displaying the total price of items in the trolley. The system eliminates traditional scanning at the counter, speeds up shopping, and reduces the possibility of human error. It is based on the Arduino platform and Xbee modules for wireless automation.
- 2) V. V., P. K. P. and C. R. S., "Smart Shopping Trolley," 2018 International Conference on Circuits and Systems in Digital Enterprise Technology (ICCSDET), 2018, pp. 1-4. Our proposed solution to reduce shopping time and improve customer experience at supermarkets involves a smart shopping trolley equipped with a barcode scanner and touchscreen display. This allows customers to scan products and view information and cost in realtime, culminating in a total bill displayed on the trolley. Payment can then be made through various online options such as Paytm, UPI, or PhonePay.
- 3) T. K. Das, A. K. Tripathy and K. Srinivasan, "A Smart Trolley for Smart Shopping," 2020 International Conference on System, Computation, Automation and Networking (ICSCAN), 2020, pp. 1-5. Our proposed solution is an IoT-based system consisting of a smart trolley equipped with an Arduino board, RFID reader, RFID tag, LCD display, ESP8266 Wi-Fi module, and a website to maintain product and customer details. The trolley can interact with a network spread worldwide, allowing customers to purchase products, receive e-bills through email, and view purchase details through the shop's website.
- 4) S. Kowshika, S. S. Madhu mitha, G. Madhu Varshini, V. Megha and K. Lakshmi, "IoT based Smart Shopping Trolley with Mobile Trolley Application," 2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS), 2021, pp. The proposed Smart Trolley in this paper offers easy and quick payment of bills using IoT and a mobile application. It uses RFID, load cell, LCD display and Raspberry Pi to scan products and prevent theft.
- 5) M. Jaishree., K. R. Lakshmi prabha., S. Jeyaprabha. and K. Mohan., "Smart Shopping Trolley Using IOT," 2021 7th International Conference on Advanced Computing and Communication Systems. This paper proposes an automated shopping trolley with a barcode scanner and LCD display, using the Raspberry Pi device. It aims to reduce waiting time in queues during rush hours at supermarkets.
- 6) M. Shahroz, M. F. Mushtaq, M. Ahmad, S. Ullah, A. Mehmood and G. S. Choi, "IoTBased Smart Shopping Trolley Using Radio Frequency Identification," in IEEE Access, vol. 8, pp. 68426-68438. The proposed IoT-based Smart Shopping Trolley utilizes RFID sensors,

SMART SHOPPING CART WITH AUTO BILLING SYSTEM

Arduino microcontroller, Bluetooth module and Mobile application to efficiently manage shopping and billing, reducing waiting time and enhancing customer experience.

Chapter 3

DESIGN CONSIDERATIONS

3.1 METHODOLOGY:

This technology offers a way to increase the pace of goods purchases and quicker payment options. In this setup, the LCD display, RFID and Arduino uno were all properly positioned inside the shopping cart. RFID tags are present on every single item. Before placing the item in the cart, an RFID reader scans the product information. The cost of the product, the total number of items, and the overall cost of the products are all displayed on the LCD display whenever clients scan the items. The consumer then hits the remove button, scans the item, which instantly lowers the item's price overall and removes it from the basket. So that the buyer must be aware of the entire number of items before proceeding to the billing area, which speeds up the purchase process. Customer can pay bill directly on the trolley. The shopping mall owners also reduce the economy on the billing counters.

3.2 BLOCK DIAGRAM

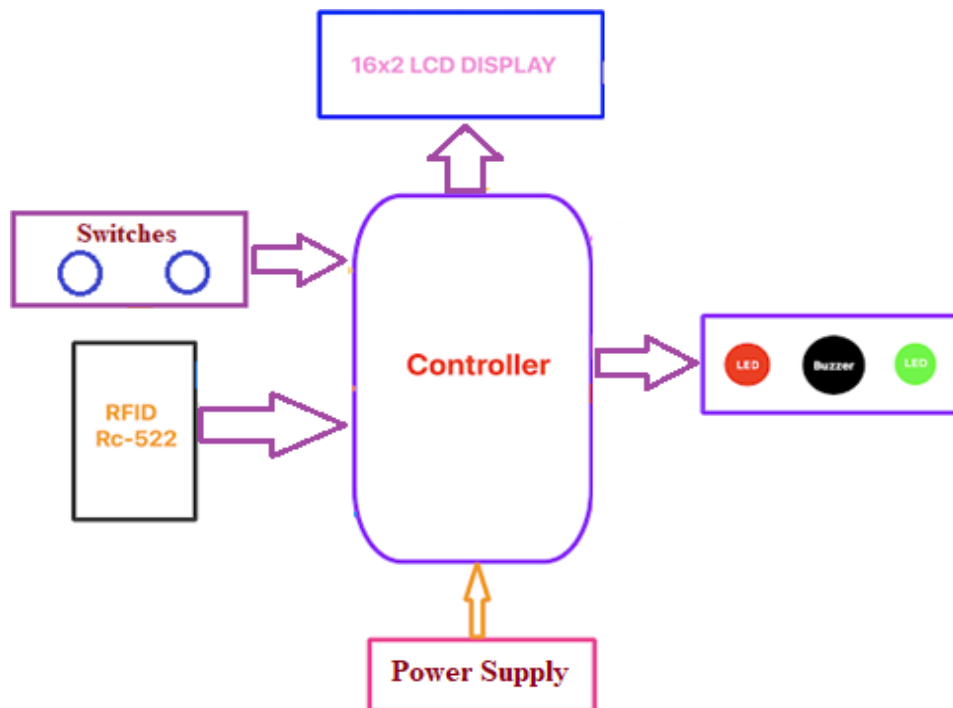


Fig 3.1 Block diagram

3.1.1 Block diagram description

The block diagram of Smart shopping cart is as shown in fig (1). The system has both hardware and software. Hardware part includes Arduino UNO, LCD display, RFID and Buzzer. Software includes the programming of Arduino UNO for RFID Scanning and Trolley movement following the customer.

The main controller of this project is Arduino Uno Board where the project code is stored. The Arduino operates all the processes from reading RFID Cards info to storing and calculating prices. The RFID Module that we are using here is the **EM-18 Module** which operates at a frequency of 125KHz. Apart from the RFID Cards, we need 3 RFID cards. These RFID cards are named as different items name like butter, milk, biscuits, etc.

The system has a buzzer that turns on when an RFID is scanned. There are two LEDs Red and Green LED. These LEDs turn on and off based on items added or removed. The most important part of the project is the LCD Display. We are using a 16x2 LCD display for this project. The 10K potentiometer is used with LCD Display for adjusting LCD contrast. When you press hold the reset button and scan the RFID, the items can be removed.

3.2 HARDWARE REQUIREMENTS

For designing this hardware many types of devices are used to make it perfectly working. All the devices are purchased from different manufacturers. These components are soldered on a soldering board. The following list of hardware is required for this system.

Components Required:

- Arduino Uno
- RFID Reader [**EDK - 125 –TTL** Module]
- RFID Cards
- 16x2 LCD
- Buzzer
- Push Buttons
- Connecting Wires
- Power supply 12v 1-amp Adapter

3.2.1 Arduino UNO R3 Microcontroller :

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega8U2 programmed as a USB-to-serial converter. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.

Technical Specification

<i>Microcontroller</i>	<i>ATmega328</i>
<i>Operating Voltage</i>	<i>5V</i>
<i>Input Voltage (recommended)</i>	<i>7-12V</i>
<i>Input Voltage (limits)</i>	<i>6-20V</i>
<i>Digital I/O Pins</i>	<i>14 (of which 6 provide PWM output)</i>
<i>Analog Input Pins</i>	<i>6</i>
<i>DC Current per I/O Pin</i>	<i>40 mA</i>
<i>DC Current for 3.3V Pin</i>	<i>50 mA</i>
<i>Flash Memory</i>	<i>32 KB of which 0.5 KB used by</i>
<i>bootloader</i>	
<i>SRAM</i>	<i>2 KB</i>
<i>EEPROM</i>	<i>1 KB</i>
<i>Clock Speed</i>	<i>16 MHz</i>

The Board

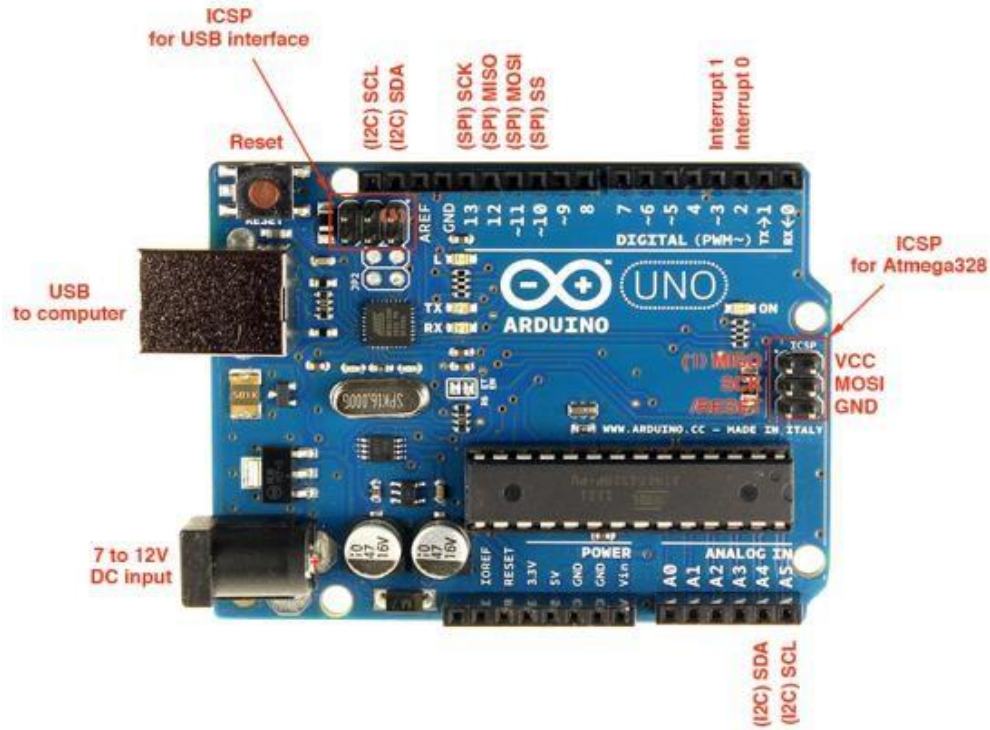


Fig 3.2 Arduino UNO Board

Power

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector. The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- VIN. The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

- 5V. The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- 3V3. A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- GND. Ground pins

Memory

The Atmega328 has 32 KB of flash memory for storing code (of which 0,5 KB is used for the bootloader), It has also 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

Input & Output :

Each of the 14 digital pins on the Uno can be used as an input or output, using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have.

SPECIALIZED FUNCTIONS:

- ❖ Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip .
- ❖ External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the `attachInterrupt()` function for details.
- ❖ PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the `analogWrite()` function.
- ❖ SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.

- ❖ LED: 13. There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though is it possible to change the upper end of their range using the AREF pin and the `analogReference()` function. Additionally, some pins have specialized functionality:

- AREF. Reference voltage for the analog inputs. Used with `analogReference()`.
- Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

COMMUNICATION

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega8U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '8U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, an *.inf file is required

The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1

A Software Serial library allows for serial communication on any of the Uno's digital pins. The ATmega328 also support I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. To use the SPI communication, please see the ATmega328 datasheet.

Pin configuration

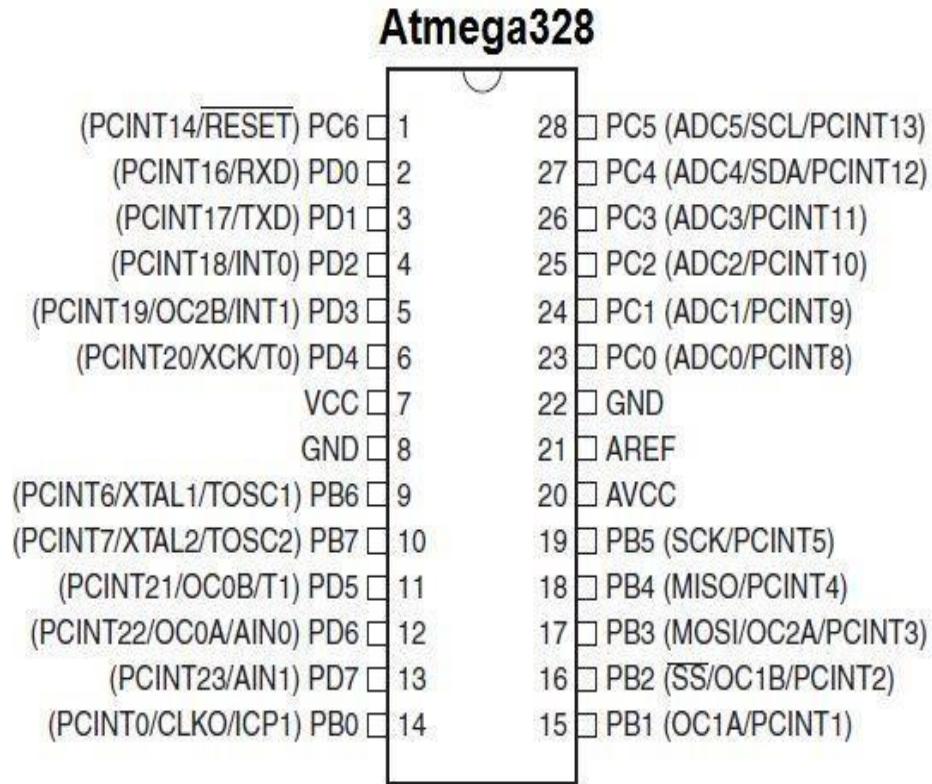


Fig 3.3 Pin configuration of the Atmega328 controller

3.2.2 RFID Reader [EDK - 125 –TTL Module]

The NSK125 series RFID Proximity OEM Reader Module has a built-in antenna in minimized form factor. It is designed to work on the industry standard carrier frequency of 125 kHz. This LF reader module with an internal or an external antenna facilitates communication with Read-Only transponders—type UNIQUE or TK5530 via the air interface. The tag data is sent to the host systems via the wired communication interface with a protocol selected from the module Both TTL and Wiegand Protocol. The LF module is best suited for applications in Access Control, Time and Attendance, Asset Management, Handheld Readers, Immobilizers, and other RFID enabled applications.

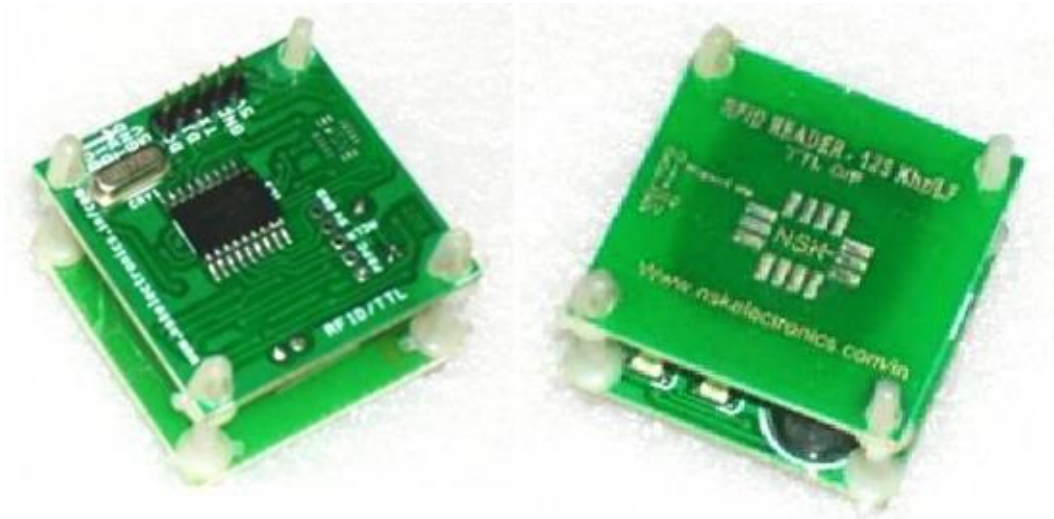


Fig 3.3 NSK125 RFID System

Components of RFID System

RFID based system has two basic components are RFID Tag & RFID Reader

RFID Tags

An RFID tag consists of an integrated circuit and an antenna. This tag carries 12 unique numbers. The tag is also composed of a protective material that holds the pieces together and shields them from various environmental conditions. The protective material depends on the application. For example, employee ID badges containing RFID tags are typically made from durable plastic, and the tag is embedded between the layers of plastic. RFID tags come in a variety of shapes and sizes. Tags can be attached to almost anything like Animals, Employee ID Card, vehicles, assets, Shoes, etc. There are two types of RFID tags.

RFID Readers

It is used to read unique IDs from RFID tags. Whenever RFID tags come in range, the RFID reader reads its unique ID and transmits it serially to the microcontroller or PC. The RFID reader has a transceiver and an antenna mounted on it. It is mostly fixed in a stationary position.

How Does RFID Works?

RFID Reader has a transceiver that generates a radio signal and transmits it through the antenna. This signal itself is in the form of energy which is used to activate and power the tag.

When the RFID tag comes in a range of signals transmitted by the reader, the transponder in the tag is hit by this signal. A tag draws power from the electromagnetic field created by the reader. Then, the transponder converts that radio signal into usable power. After getting power, the transponder sends all the information it has stored in it, such as a unique ID to the RFID reader in the form of an RF signal. Then, the RFID reader puts this unique ID data in the form of a byte on a serial Tx (transmit) pin. This data can be used or accessed by PC or microcontroller serially using UART communication.

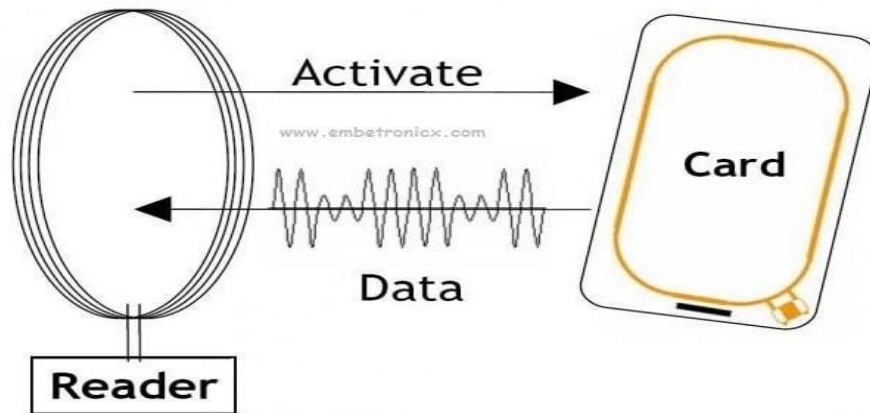


Fig 3.4 Working of RFID System

Applications

RFID is used in many applications like an attendance system in which every person will have their separate RFID tag which will help identify the person and their attendance.

RFID is used in many companies to provide access to their authorized employees.

It is also helpful to keep track of goods and in an automated toll collection system on the highway by embedding Tag (having unique ID) on them.

3.2.3 I2C LCD display

If you've ever attempted to connect an LCD display to an Arduino, you've probably noticed that it uses a lot of Arduino pins. Even in 4-bit mode, the Arduino requires seven connections – half of the Arduino's available digital I/O pins.

The solution is to use an I2C LCD display. It only uses two I/O pins that are not even part of the digital I/O pin set and can be shared with other I2C devices.

Hardware Overview

A typical I2C LCD display consists of an HD44780-based character LCD display and an I2C LCD adapter. Let's learn more about them.

Character LCD Display

As the name suggests, these LCDs are ideal for displaying only characters. A 16×2 character LCD, for example, can display 32 ASCII characters across two rows.

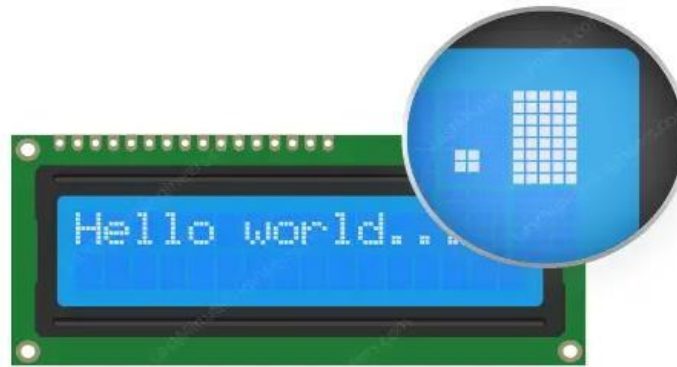


Fig 3.5 LCD Matrix

If you look closely, you can see tiny rectangles for each character on the screen as well as the pixels that make up a character. Each of these rectangles is a grid of 5×8 pixels.

I2C LCD Adapter

At the heart of the adapter is an 8-bit I/O expander chip – PCF8574. This chip converts the I2C data from an Arduino into the parallel data required for an LCD display.



The board also includes a tiny trimpot for making precise adjustments to the display's contrast.



There is a jumper on the board that provides power to the backlight. To control the intensity of the backlight, you can remove the jumper and apply external voltage to the header pin labeled 'LED'.

I2C Address of LCD

If you have multiple devices on the same I2C bus, you may need to set a different I2C address for the LCD adapter to avoid conflicting with another I2C device.

For this purpose, the adapter comes with three solder jumpers/pads (A0, A1, and A2). The address is set when a jumper is shorted with a blob of solder.

All three address inputs are pulled HIGH using onboard pullups. This gives the PCF8574 a default I2C address of 0x27 or 0x3F.

I2C LCD Display Pinout

The I2C LCD Display has only four pins. The following is the pinout:



Fig 3.6 Pin outs of I2C LCD

GND is a ground pin.

VCC is the power supply pin. Connect it to the 5V output of the Arduino or an external 5V power supply.

SDA is the I2C data pin.

SCL is the I2C clock pin.

Wiring an I2C LCD Display to an Arduino

Connecting an I2C LCD is much simpler than connecting a standard LCD. You only need to connect four pins.

Begin by connecting the VCC pin to the Arduino's 5V output and the GND pin to ground.

Now we are left with the pins that are used for I2C communication. Note that each Arduino board has different I2C pins that must be connected correctly. On Arduino boards with the R3 layout, the SDA (data line) and SCL (clock line) are on the pin headers close to the AREF pin. They are also referred to as A5 (SCL) and A4 (SDA).

3.2.4 Buzzer :

An Active Buzzer Alarm Module easily interfaced with an Arduino or other microcontrollers is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Just like what you are viewing now, it is 3.3V to 5V DC power supply Electronic Part Active Buzzer Module. Using top quality material, it is long durable in use.

An Active buzzer rings out as long as it is electrified. Compared with a passive buzzer, it is a bit expensive but easier to control. Typical uses of buzzers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke.



Fig 3.7 Pin outs of I2C LCD

Specifications: Active Buzzer Module

- Product Name: 3.3 to 5V Active Buzzer Alarm Module Sensor
- Transistor drive module uses 8550

- With fixed bolt hole- easy installation- 2.6mm aperture.
- Operating voltage 3.3V-5V

Active buzzer features are:

- An active buzzer with a concussion internal source- so long as it will be called an energized
- Program easy to control- SCM can let a high-low sound- while passive buzzer did not

Applications:

1. Fire Alarm
2. Audio signaling device
3. Alarms
4. Security System
5. Timers

3.2.5 Pushbutton Switch:

Pushbutton Switches operate via a pushing or pulling action that moves the operating part in the direction of the force required to open or close the contacts. The operating part is generally equipped with an incandescent lamp or LED to provide illumination and status indication.



Fig 3.8 Pushbutton Switch

3.2.6 12V-1Amp DC Adapter:

This is a high quality SMPS based 12 Volt DC Adaptor. It can be used for any device that requires 12V DC supply like led strip lights, Fingerprint scanners, RFID readers, GSM Modems, GPS Modules, etc.

Features of 12V/1A Adapter

- Input Voltage : 100 - 230 Volts.
- Frequency : 50/60 Hz
- Output Current : 1 Ampere
- Connector Size : 5.5 mm (Outside barrel) x 2.5 mm (Inside Barrel)
- Center : Positive.

3.3 SOFTWARE REQUIREMENTS

3.3.1 Arduino Ide

The Arduino hardware and software was designed for artists, designers, hobbyists, hackers, newbie's, and anyone interested in creating interactive objects or environments. Arduino can interact with buttons, LEDs, motors, speakers, GPS units, cameras, the internet, and even your smart-phone. This flexibility combined with the fact that the Arduino software is free, the hardware boards are pretty cheap, and both the software and hardware are easy to learn has led to a large community of users who have contributed code and released instructions for a huge variety of Arduino-based projects. Here we set the meteorological factors value initially through the arduino programming and if the value crossed ahead of this value, than buzzer will sound than we come to know the respectively meteorological factors values. Therefore we can continuously monitor and control of these values.

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while

saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the **File > Sketchbook** menu or from the **Open** button on the toolbar. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the **Preferences** dialog

When you upload a sketch, you're using the Arduino **boot loader**, a small program that has been loaded on to the microcontroller on your board. It allows you to upload code without using any additional hardware. The boot loader is active for a few seconds when the board resets; then it starts whichever sketch was most recently uploaded to the microcontroller. The boot loader will blink the on-board (pin 13) when it starts (i.e. when the board resets).

3.3.2 Embedded C

Embedded C is a set of language extensions for the C Programming language. C is often used for system programming, including implementing applications. Embedded C uses most of the syntax of standard C, e.g., `main ()` function, variable definition, data type declaration, conditional statements (if, switch, case), loops (while, for), functions, arrays and strings, structures etc.

It is small and simpler to learn, understand, program and debug. It is efficient & supports access to I/O and provides ease of management of large embedded projects.

The compiler derives its name from the way it works, looking at the entire piece of source code and collecting and reorganizing the instruction. See there is a bit little difference between compiler and an interpreter. Interpreter just interprets whole program at a time while compiler analyzes and execute each line of source code in succession, without looking at the entire program.

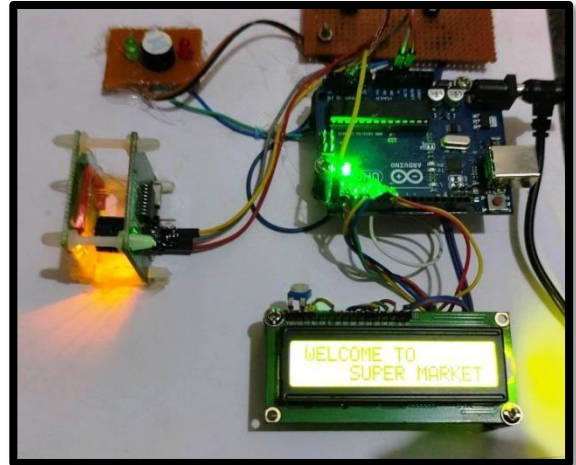
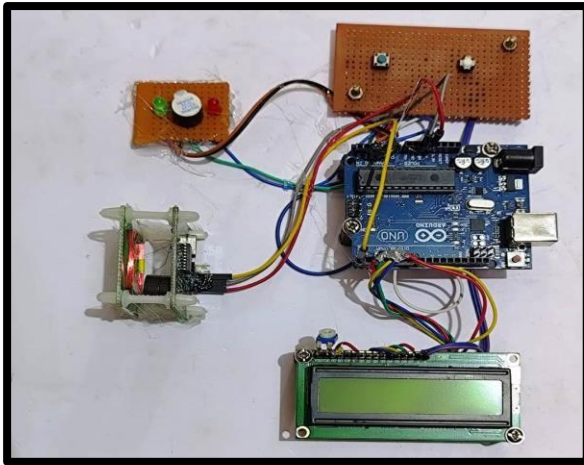
Chapter 4

RESULTS & DISCUSSIONS

All the experimental parts were discussed above, in which proposed system LCD is used to display the product information in terms of cost. The proposed model is easily accessible, does not require special training and convenient to use. The manpower is decreased and will save time that the user spends in billing queue. Many users can be attended in same time which is useful for retailers and customers. Time efficiency and cost efficiency are guaranteed by this smart billing system. Smart trolley envisions simplifying the billing procedure by facilitating customers in creating a shopping session which lasts until the customer commands it to be cleared and preserves the data of each product in the basket by using RFID tags to make the entry. It also facilitates in keeping the shopping experience in budget by displaying the total cost to the customer. Emerging trend of online shopping, which cuts down the hassle, at the same time that shopping at stores introduction of environment friendly smart carts and smart baskets not only help the stores to eliminate the surge but also serve to reduce the usage of paper, the number of employees making it more economical and unnecessarily wasted in printing copies of bills. Thus the use of RFID based smart trolleys is of major significance or importance interest and welfare to the society.

The results of shopping done by customers are shown on Liquid Crystal Display in terms of cost, item name and in some cases, display will guide you that product is expired where customers have to be cautious. As the RFID Reader Scans the items from RFID tag the Item name along with cost are displayed this process continues as when items are added and amount of items is cumulative in process. As the items are removed the amount will be subtracted due to item may be disliked.

SMART SHOPPING CART WITH AUTO BILLING SYSTEM



Chapter 5

APPLICATIONS, ADVANTAGES & DISADVANTAGES

5.1 APPLICATIONS

- **Supermarkets and Hypermarkets:** This system is ideal for large stores where customers typically spend significant amounts of time shopping, helping to speed up the entire process.
- **Retail Stores:** It can be used in fashion, electronics, or general retail stores, where customers can quickly check out without waiting in line.
- **E-commerce Integration:** Future versions of smart trolleys could integrate with e-commerce platforms, enabling customers to add items they are considering online, allowing seamless transfer between in-store and online shopping.

5.2 ADVANTAGES

- ✓ **Improved shopping experience:** Smart shopping carts can make shopping more efficient and enjoyable.
- ✓ **Time savings:** Smart shopping carts can help customers save time by automatically scanning items and eliminating the need to wait in line.
- ✓ **Cost savings for retailers:** Smart shopping carts can help retailers reduce labor costs by reducing the number of cashiers needed.
- ✓ **Better Store Management:** Real-time data collection on inventory and customer shopping behavior can help retailers manage stock levels and optimize their sales strategies.
- ✓ **Sustainability:** Smart shopping carts can help reduce paper waste and retailers can use the data collected to reduce their carbon footprint.

5.3 DISADVANTAGES

- **Cost of Implementation:** Developing and maintaining smart trolleys can be expensive, especially for smaller retailers who may not have the budget for such advanced systems.

- **Privacy Concerns:** Collecting data on shopping patterns and linking transactions to customer identities may raise privacy concerns. Retailers must ensure data protection measures are in place.
- **Technological Limitations:** The technology is still evolving, and there may be limitations in certain environments, such as handling complex products, mixed bundles, or products that lack RFID or barcode labeling.
- **Customer Adaptability:** Some customers may find it challenging to adopt new technology, especially older shoppers or those less familiar with mobile payment options.

Chapter 6

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

“Smart Shopping Cart using Arduino and RFID” has been successfully implemented. This system is not only effective in eradicating the long queues but also manages the budget of the customer. This system is automated and far better than the existing Barcode system. With new technologies rapidly making every walk of life smart, shopping should be made smarter too. The system also has a very quick and easy billing option.

6.2 FUTURE SCOPE

The transferring of information from the trolley/basket to the Admin's system can be made wireless by adopting IoT. Also, with emerging technologies, the movement of the cart can be automated, too. Hence, this system has a number of future applications and can be the basis of some advanced inventions in the future

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